

The Accuracy Illusion: How Overlapping Samples Manufacture Confidence, Not Skill, in Intraday Index Forecasting

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ABSTRACT

Intraday forecasting systems routinely report directional hit-rates between 60% and 85%, and those figures move real capital. We show they are not valid estimates of skill. A k -bar-ahead forecast issued every bar shares $k-1$ of its k outcome bars with its neighbour; when the model's calls also persist, as they do on a live system we instrument, the reported hit-rate acquires a variance far larger than its nominal sample size implies. The reading is not biased upward. Its stated precision is. A forecaster with *exactly zero skill*, simulated against real NSE index bars, posts a $\geq 71\%$ session once in 50 while reporting an interval $1.66\times$ too narrow, rising to $3.11\times$ at the per-minute density these systems publish.

We give the closed form behind this, the classical design effect with clusters given by runs of identical calls, and a calibrated interval built on it that covers at 96% where the conventional one covers at 75%. Applied to our own results it leaves nothing: across 48 model-index-horizon tests, 16 clear $p < 0.05$ scored per bar and zero clear it per event or at an effective sample size. A premium-selling strategy we had published as this project's one measurable edge fails the same correction. Break-even accuracy derived from measured move distributions is 71.5% at thirty minutes against a best achieved 51.2%, and at five minutes no accuracy suffices under a symmetric bracket. Code and raw results are public.

KEYWORDS

overlapping observations, backtest overfitting, directional forecasting, negative results, reproducibility, index derivatives

1 INTRODUCTION

A system reporting 71% directional accuracy is, on its face, extraordinary. Work that controls for scoring unit, costs and multiple testing places realistic out-of-sample intraday index accuracy close to a coinflip [10, 17], while surveys of the applied literature catalogue many higher figures reported without those controls [27].

This paper began as an attempt to build such a system. Over fourteen months we ran a live multi-horizon forecasting platform for Indian index derivatives, and it reported 71%. That number was wrong. The specific session producing it cannot be re-audited, because the portion of the log holding it was lost before this analysis; every figure below comes from sessions that survive.

The econometrics of overlapping observations is not new [1, 2]. Boudoukh et al. [3] show that overlap manufactures apparent long-horizon predictability which vanishes under correct inference. What that literature does not document is the magnitude of the distortion in a now-ubiquitous practice: the every-minute multi-horizon

dashboard, where overlap meets a second ingredient, prediction persistence, and is then amplified across a panel.

Contributions. (i) We measure the artefact as it occurs, showing that overlap alone changes nothing and that inflation requires overlap *and* persistent predictions, with persistence calibrated from a production log. (ii) We derive the effective sample size of such a stream, show it is governed by $\mathbb{E}[L^2]/\mathbb{E}[L]$ over run lengths, and calibrate an interval that covers at its nominal rate. (iii) We apply it to our own results and to a complete control set — shuffled labels, injected positive controls, nine architectures, a block-bootstrap null — and report that nothing survives. (iv) We invert accuracy into a required-versus-achieved framing that shows the gap is not close. Code, raw results and the free-data corpus are at <https://github.com/rachitranka25/accuracy-illusion>.

2 RELATED WORK

Hansen and Hodrick [1] and Richardson and Smith [2] characterise inference under overlapping returns; Newey and West [4] give the standard correction. That work concerns regression coefficients. We apply the same logic to what practitioners publish, the session hit-rate, and the correction we reach is the classical design effect [5, 6] with clusters given by runs of identical calls. The contribution is its application to forecast streams, not the statistic.

On multiplicity, Harvey et al. [13] argue the conventional $t > 2$ threshold is indefensible after decades of factor mining. Bailey and López de Prado [14] introduce the Deflated Sharpe Ratio and Bailey et al. [15] the Probability of Backtest Overfitting; White's Reality Check [11], its refinement by Hansen [12] and Benjamini-Hochberg [9] provide complementary controls. Sullivan et al. [10] apply the Reality Check to a universe of trading rules, the direct precedent for our strategy sweep.

On machine learning for finance, Fischer and Krauss [25] and Krauss et al. [26] report headline performance gross of costs; Gu et al. [24] give a comprehensive comparison with modest out-of-sample R^2 . Zhang et al. [23] and Kolm et al. [22] demonstrate genuine short-horizon alpha, both from message-level exchange data rather than bars — the distinction that matters here.

3 DATA AND METHOD

All data is obtainable free from a retail brokerage account and public exchange archives (Table 2). Three practical observations. Index spot prints zero volume, so a feature computed as a "VWAP" on spot is an unweighted expanding mean; real volume exists only on futures. The exchange's *live* option-chain API is geo-restricted while its static end-of-day archive is not, which turned an assumed constraint into a decade of data. And brokers authenticating by

Table 1: Methodological comparison with representative prior work. “Scoring unit” is the unit in which hit-rate or return is aggregated; per-bar scoring of a multi-bar horizon is the condition under which the artefact of Section 4 appears. MT: multiple-testing correction. NC: null control (shuffled labels or equivalent). C: transaction costs modelled. R: code and raw results public.

Study	Market	Data granularity	Scoring unit	MT	NC	C	R
Fischer & Krauss [25]	S&P 500	Daily bars	Per day, non-overlapping	—	—	pre-cost	—
Krauss et al. [26]	S&P 500	Daily bars	Per day, non-overlapping	—	—	pre-cost	—
Gu et al. [24]	US equities	Monthly	Per month	partial	—	partial	partial
Zhang et al. [23]	LSE	Message-level LOB	Per event	—	—	—	partial
Kolm et al. [22]	Nasdaq	Message-level LOB	Per horizon	—	—	discussed	—
This work	NSE index	5-min bars, EOD chains	Per event (k-bar)	✓	✓	✓	✓

MT here means Bonferroni and Benjamini–Hochberg, which applied cleanly, plus the Deflated Sharpe Ratio and PBO/CSCV, which we ran and found inapplicable and degenerate respectively (Section 6). NC means shuffled-label controls on every model plus injected positive controls. C means explicit round-trip costs, inverted into a required-accuracy threshold. We omit a row for “typical applied studies”: characterising that population would require citing specific papers as negative examples, and the argument does not need it.

Table 2: Free-data corpus. All collectors are public.

Asset	Period	Size
BANKNIFTY 5-min bars	2025-02 – 2026-07	26 265
NIFTY 50 5-min bars	2025-02 – 2026-07	26 267
3 further indices, 5-min	2025 – 2026	9.6k–14k
Option chain EOD + OI	2024-07 – 2026-07	1 382 738
Long-history option chain	2016-05 – 2026-05	2 181 d
Participant-wise OI	2024-07 – 2026-07	515 d
Index futures (real volume)	2024-01 – 2026-07	639 d

locally computed TOTP permit collection from any jurisdiction, where SMS-based flows do not.

Every accuracy passes through one harness. Splits are expanding-window walk-forward with the k bars whose labels overlap the test window purged from training [16]. Accuracies carry Clopper–Pearson [34] intervals, exact binomial p -values and the Pesaran–Timmermann statistic [8], which is the appropriate null for directional accuracy. Every model is additionally trained on randomly permuted labels; if real and shuffled scores coincide, the pipeline is reading noise. A synthetic feature of known strength is injected as a positive control, since a pipeline that cannot recover planted signal cannot be trusted to report a null. Strategy returns use non-overlapping k -bar holds with explicit costs.

Units. All analysis is on 5-minute bars, so the 30-minute horizon is $k = 6$ and a session yields $n = 69$ forecasts. The production system emits a forecast every *minute*; pooled across indices its run lengths average 13.5 minutes (3.0 bars) with a longest of 207 minutes (41 bars).

4 THE MECHANISM

4.1 Overlap is arithmetic

Figure 1 shows the autocorrelation of the k -bar forward return two ways. On overlapping windows it reaches 0.841 at lag 1 and decays to approximately zero at *exactly* lag k ; sampled so no two observations share a bar it is 0.067 and indistinguishable from zero thereafter. The pattern reproduces at $k = 6$ and $k = 12$, with the decay landing on lag 6 and lag 12. The correlation is a property of the sampling scheme, not the market.

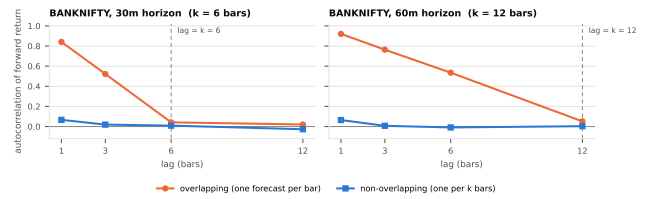


Figure 1: Forward-return autocorrelation, overlapping versus non-overlapping, at two horizons. The overlapping series dies at exactly lag k in both panels.

Table 3: A forecaster with exactly zero skill, 30-minute horizon, scored per bar versus per event. BANKNIFTY; NIFTY 50 within 0.3 points on every row.

	Per bar	Per k bars
Reported sample size per session	69	12
Stated binomial 95% CI	± 11.8 pts	± 28.3 pts
Realised spread across sessions	± 19.6 pts	± 28.0 pts
Understatement factor	1.66×	0.99×
$P(\text{session} \geq 60\%)$	15.7%	19.2%
$P(\text{session} \geq 71\%)$	2.0%	7.1%

4.2 Persistence is the second ingredient

Overlap alone is insufficient. If predictions are drawn independently each bar, the hit indicator is i.i.d. Bernoulli regardless of how autocorrelated the outcome is. Our first simulation made exactly this error and correctly found nothing. Real models do not re-decide every bar: measured on the production log at the 30-minute horizon, BANKNIFTY calls persist for a mean of 15.2 minutes and NIFTY 50 for 12.2, against a mean run of one for a model that re-drew each bar. When both the call and the outcome persist, one trending stretch is recorded as dozens of hits.

4.3 What a zero-skill forecaster reports

We simulate a fair coin held for a run length drawn from that distribution, evaluated against real bars and scored as a dashboard scores it (Table 3).

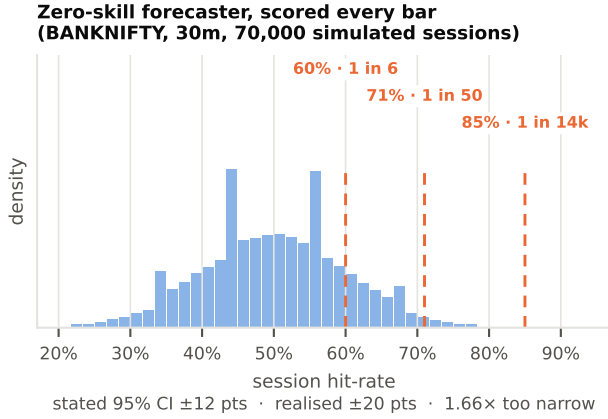


Figure 2: Measured session hit-rate distribution of a zero-skill forecaster scored every bar. The multimodality is not noise: with roughly a dozen independent decisions per session the hit-rate lands on near-discrete values, which is the small effective sample made visible.

Figure 2 shows the distribution the table summarises. The table states a point easily read backwards. Per-bar scoring does *not* produce extreme readings more often than honest scoring; it produces them *less* often. What it does is report them with an interval $1.66\times$ too narrow, because the quoted n of 69 is a fiction. Overlap manufactures unwarranted *confidence*, not high hit-rates, and it is that false precision which misleads.

4.4 At the real granularity, and across a panel

The production system forecasts every minute, so a session yields about 314 scored forecasts rather than 69. Taking the outcome series from the log itself and simulating the same zero-skill forecaster, the understatement factor rises to $3.11\times$ while extreme sessions become slightly *rarer*. Every 5-minute-bar figure here therefore understates the artefact.

A dashboard also watches more than one number: two indices at four horizons is eight hit-rates per session. Simulating the panel jointly on the same sessions gives $P(\exists \text{ series} \geq 71\%) = 15.1\%$, against 15.2% under an independence assumption — adequate here because a zero-skill forecaster’s hits are near-independent across series even when outcomes are not. Either way, about one session in seven.

5 A CALIBRATED ESTIMATOR

5.1 Effective sample size

Let $h_t = 1\{\hat{y}_t = y_t\}$ and $\hat{p} = n^{-1} \sum_t h_t$. Practitioners quote $\text{Var}(\hat{p}) = p(1-p)/n$, valid only for independent h_t . In general

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n} \left[1 + 2 \sum_{j=1}^{n-1} \left(1 - \frac{j}{n}\right) \rho_j \right], \quad (1)$$

with $\rho_j = \text{Corr}(h_t, h_{t+j})$. Because the dependence here is block-shaped, this collapses. If the sequence decomposes into runs of

Table 4: Coverage of a nominal 95% interval on a session hit-rate. BANKNIFTY, 30-minute horizon, nominal $n = 69$; NIFTY 50 agrees within 2 points. HAC-L gives the Bartlett kernel a run-structure bandwidth; *calibrated* divides the design effect by two.

Skill	Naive	HAC	HAC-L	Design	Calib.	n_{eff}
50%	75.5	87.0	88.4	99.5	96.0	12.4
55%	74.8	86.6	88.2	99.6	96.1	12.0
59%	74.0	85.6	87.0	99.6	95.8	11.2

length L_i within which h_t is constant, then

$$n_{\text{eff}} = \frac{n}{\sum_i L_i^2/n} \xrightarrow{n \rightarrow \infty} \frac{n \mathbb{E}[L]}{\mathbb{E}[L^2]}. \quad (2)$$

This is the classical design effect [5, 6]; what is new is the observation that a forecast stream has exactly this structure and that nobody reports it. The inflation is governed by $\mathbb{E}[L^2]$, so the tail matters, not the mean.

5.2 Validation by coverage

Table 4 reports coverage against a forecaster of known skill whose calls persist per the measured distribution, with true skill measured from the generative process rather than assumed.

HAC and a block bootstrap (86.6%, omitted for space) both under-cover by about eight points. The obvious diagnosis is bandwidth: the automatic Bartlett rule gives 3.7 bars at $n = 69$, close to the pooled mean run of 3.0, while the tail reaches 41. We tested that rather than asserting it — a run-structure bandwidth recovers one to two points and no more, so bandwidth was part of the problem and not the whole of it, and we report the repair though it mostly fails. The design effect over-corrects instead, because treating runs as independent clusters ignores that consecutive runs are negatively dependent. The over-statement is close to a constant factor of two, visible both from coverage and from the realised spread in Table 3, which implies $n_{\text{eff}} \approx 25$ where (2) gives 12. Dividing by that factor yields the calibrated interval, at 95.8–96.1% across skill levels and both indices.

5.3 Reliability, and a real session

Equation (2) depends on a tail-driven second moment estimated from a few dozen runs. Measured against a pooled reference of 11.0, a single session’s n_{eff} lands between $0.65\times$ and $1.67\times$ nine times in ten. So n_{eff} should be estimated from a run of sessions, and treated as an order of magnitude from one.

That matters immediately. Applying the estimators to the production log, a BANKNIFTY session reporting 57.1% has a naive interval excluding 50% and a corrected interval of $[34.4, 79.9]$. The best NIFTY 50 session, 69.6% over 335 forecasts, has a corrected interval of $[50.0, 89.1]$ — which excludes 50%, but at the 5th percentile of the n_{eff} uncertainty becomes $[45.3, 93.8]$ and does not. The lower bound is smaller than the estimator’s own noise, so the session supports no claim. An earlier draft reported that it did.

Per-event scoring is unbiased and needs no estimator, and is what we use for every backtest here. It is insufficient for a live

Table 5: Model-independent ceiling test, 30-minute horizon, five indices. Training accuracy is 100.0% on every row. Positive controls have recoverable accuracies of about 55% and 65% by construction.

Index	n	Real	Shuf.	Maj.	$s=.10$	$s=.30$
BANKNIFTY	16 856	50.5	50.2	49.3	52.4	63.5
NIFTY 50	16 853	50.0	50.2	49.8	52.0	63.7
FINNIFTY	8 988	50.2	50.5	49.4	52.4	62.5
MIDCPNIFTY	8 309	50.2	50.8	49.7	53.3	62.0
SENSEX	6 144	49.7	50.1	49.0	52.5	62.1

Table 6: Model families significant at $p < 0.05$, by horizon and scoring rule. Eight families per cell, excluding the majority-class baseline.

Horizon, index	Per bar	At n_{eff}	Per event
15 m BANKNIFTY	0	0	0
15 m NIFTY 50	2	0	0
30 m BANKNIFTY	3	0	0
30 m NIFTY 50	1	0	0
60 m BANKNIFTY	5	0	0
60 m NIFTY 50	5	0	0
Total (48 tests)	16	0	0

stream for three reasons: a dashboard cannot wait k bars or discard forecasts already shown; subsampling throws away 83% of the observations; and which of the k offsets is chosen moves the same session by 7 to 12 points, so an analyst free to pick the offset is free to pick the answer.

6 PREDICTABILITY UNDER CORRECT SCORING

6.1 Is there anything to predict?

An unconstrained forest was fitted until it memorised the training data, on five indices across two exchanges (Table 5). Each memorises completely and generalises at chance; permuting labels moves accuracy by 0.3 to 0.7 points; no real-label result is distinguishable from a coinflip ($p = 0.22$ to 0.91). Both positive controls are recovered everywhere at $p < 0.001$, so the pipeline detects signal whenever signal is present.

6.2 Nine model families, four horizons

Nine families were trained on identical splits, features, seeds and controls: logistic regression, gradient boosting, LightGBM [32], XGBoost [31], a Breiman forest [33], k -nearest neighbours, an LSTM [29], a Transformer [30] and a compact Temporal Fusion Transformer [28]. Table 6 scores every one three ways.

Scored per bar, XGBoost reaches 51.3% on BANKNIFTY at $p = 0.001$. That was, in an earlier draft, our one surviving positive result, described as Bonferroni-robust. It does not survive our own correction: re-scored one observation per 6 bars it is 50.4% at $p = 0.692$, and widening intervals to n_{eff} puts 50% inside every one.

Table 7: Break-even accuracy at 4 bp round-trip cost. p_{med}^* assumes a symmetric bracket at the median move; p_{mean}^* is the requirement for a hold-to-horizon rule. Achieved is the best per-event accuracy of nine families.

Horizon	Med.	Mean	p_{med}^*	p_{mean}^*	Ach.
<i>BANKNIFTY</i>					
5 min	22.1	30.8	>100	86.4	—
15 min	37.7	52.9	79.8	71.3	51.3
30 min	52.3	73.8	71.5	65.3	51.1
60 min	72.9	103.8	65.4	60.8	52.2
<i>NIFTY 50</i>					
5 min	8.1	11.5	>100	93.0	—
15 min	14.1	19.8	84.9	74.9	50.5
30 min	19.7	27.8	75.1	67.8	51.2
60 min	27.7	38.5	67.7	62.7	50.1

Across all 48 tests the count falls from 16 to zero. The count of apparent discoveries rises with horizon, from none at 15 minutes to five at 60 on both indices, exactly as it should if the artefact scales with k ; the count of surviving discoveries stays at zero.

Table 6 also contains its own diagnostic. The NIFTY 50 Transformer sits at 48.3% with $p < 0.001$, which at the nominal n is 4.4 standard errors below chance where the largest $|z|$ across eighteen tests should land near 2.6–2.9. That is not a broken model; it is the variance inflation surfacing inside our own results, and at n_{eff} the anomaly vanishes.

6.3 Eighty-six strategies

Separately, 86 parameterised rules — momentum, mean reversion, moving-average crosses, RSI, Bollinger, breakout, gap, volatility regime, time-of-day and streak variants — were evaluated on non-overlapping 30-minute holds at 4 bp cost with a one-sided test for profit. Two have a positive Sharpe; none is significantly profitable, before or after Benjamini–Hochberg. The Deflated Sharpe Ratio returns an implausible null benchmark of 7.77 here, because trial Sharpe dispersion is dominated by deterministic turnover differences rather than noise, so we replaced it with a circular block bootstrap [7] that resamples forward returns while holding each strategy’s positions fixed. The best observed Sharpe is 0.21 against a null best-of-86 median of 0.44, with $P(\text{null} \geq \text{observed}) = 0.658$: the winner is worse than what selection alone produces. PBO via CSCV [15] is degenerate here and reported as such.

7 ECONOMIC SIGNIFICANCE

For a symmetric bracket with target and stop at the typical move m and round-trip cost c , expectancy is $m(2p - 1) - c$, so break-even requires

$$p^* = \frac{1}{2} + \frac{c}{2m}. \quad (3)$$

The ratio c/m is the entire problem (Table 7).

Two consequences (Figure 3). At the shortest horizon the median move is smaller than the round trip, so under a symmetric bracket p^* exceeds 100% and no accuracy suffices. That conclusion is specific to the bracket: moves are strongly right-skewed, so a hold-to-horizon rule requires 86% on BANKNIFTY and 93% on

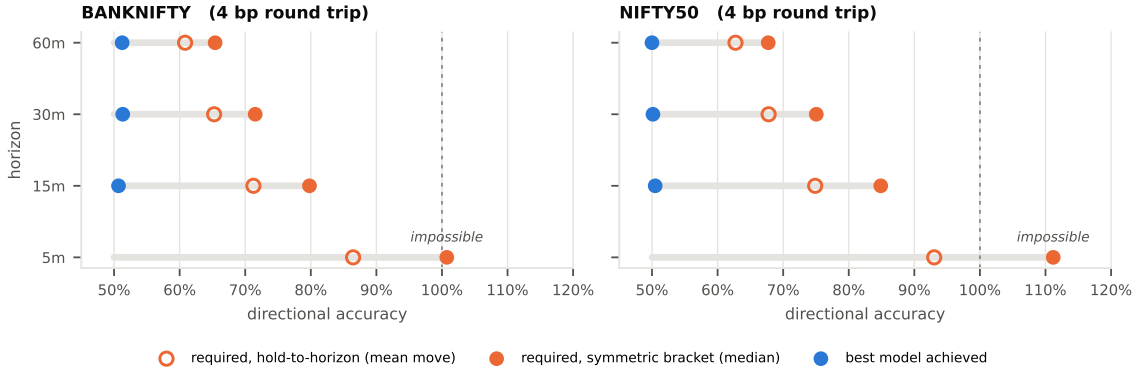


Figure 3: Break-even accuracy against the best per-event accuracy any model family reached, by horizon and index, under both payoff models. Points beyond the dashed line at 100% are horizons at which no directional accuracy is sufficient under a symmetric bracket.

NIFTY 50 instead — not impossible, merely far beyond anything here or in the literature. We report both, and the point survives either way.

Second, instrument choice can make the requirement unreachable. At 4 bp BANKNIFTY needs 71.5% at thirty minutes; at 8 bp, 93.0%; at the 20 bp a retail ATM option round trip costs over that horizon, $c > m$ and $p^* > 1$. The instrument most retail participants use, over the horizon they hold, has no directional accuracy that makes it profitable.

This also reframes the claims the paper is about. A genuine 85% would clear break-even by 13.5 points on BANKNIFTY futures and remain unprofitable on options. Such a claim is therefore either an artefact of Section 4 or true and far less valuable than it sounds.

8 WHAT IS PREDICTABLE INSTEAD

These results went through the same harness, and what it leaves is narrow enough to state precisely: *none of this project’s findings survives as a finding of ours.*

Volatility. A binary magnitude target reaches 70.8% per bar and 71.3% per event on BANKNIFTY (72.0 and 72.1 on NIFTY 50), with shuffled controls at 52–54%. That it is untouched by the correction which destroyed the directional result matters, since an instrument flattening everything would be useless. But a persistence rule already scores 67.4%, the model’s edge is +3.3 to +3.7 points per event, and at n_{eff} the model and baseline intervals overlap. This is a confirmation of volatility clustering [18, 19], not a discovery of ours.

Variance risk premium, withdrawn. Short ATM straddles held to expiry, on real end-of-day premiums, across four entry points: no configuration is significantly profitable on either index, before or after Benjamini–Hochberg, and every bootstrap interval on the mean contains zero. Recentering each configuration and re-sampling gives a best-of-four null median of 17.8 against an observed 12.1 on NIFTY ($P = 0.68$). NSE also moved BANKNIFTY options from weekly to monthly expiry mid-sample, so the two indices were never comparable: split on expiry spacing, the weekly sub-sample returns –67.5 points per trade and the monthly +57.5.

The premium is real and documented [20, 21]; our measurement of it as an edge is withdrawn.

Option-chain positioning. Over 2181 days, after regressing out the price component, six of eight residual features clear $p < 0.05$ uncorrected and *one* survives Benjamini–Hochberg: the daily change in put–call open-interest ratio ($IC = 0.059$, $t \approx 2.8$, $\approx 51\%$ hit-rate), which is also the only feature strengthening in the second half. Flow beats position. Nothing in the price-only set clears $p < 0.05$ at all.

9 LIMITATIONS

The calibration constant is fixed at 2.0 rather than derived, validated only by coverage, on one data-generating process. Its input is imprecisely measured from a single session. Two indices carry the primary intraday analysis over seventeen months in one market; the ceiling test replicates across five but the model zoo and strategy sweep do not. Break-even is computed for two payoff models, neither of which models the path, so the probability of touching a stop before a target is unaccounted. Some figures in our research record come from scripts not retained; two that were re-run did not reproduce and are withdrawn in place. Index spot carries no volume, so VWAP-derived features are misnamed. Our Temporal Fusion Transformer is a compact variant and should not be read as the published architecture [28]. Finally, the order-flow hypothesis, where the literature locates genuine short-horizon alpha [22, 23], is untested here for want of history.

10 CONCLUSION

The headline metric of intraday directional forecasting is not merely noisy. Its stated precision is systematically wrong, by enough to make a coinflip look like a discovery: a zero-skill forecaster posts a $\geq 71\%$ session once in 50, and once every seven sessions somewhere on a modest panel, while reporting an interval $1.66\times$ too narrow at 5-minute granularity and $3.11\times$ too narrow at the density these systems publish.

Applied to our own work the correction leaves nothing standing. Sixteen of 48 model–index–horizon tests clear $p < 0.05$ per

bar and none clears it per event. Eighty-six strategies produce a winner worse than a bootstrap null. Premium selling, which we had published as this project's one measurable edge, fails the same correction. The one untouched result confirms volatility clustering rather than finding anything.

So the honest summary is that this project found nothing tradeable, and its value lies in how thoroughly that was established. What we offer is a method: an effective sample size for overlapping persistent forecasts, a calibrated interval built on it, and a framing that turns an accuracy figure into a question about cost. Reporting practice should change to match, and the change is cheap. Scoring unit, effective sample size, multiplicity correction, a shuffled-label control and an explicit cost model are not optional refinements to an accuracy figure. Without them the figure does not mean anything, and we know that because we published several that did not.

REFERENCES

- [1] L. P. Hansen and R. J. Hodrick. 1980. Forward exchange rates as optimal predictors of future spot rates. *J. Political Economy* 88, 5 (1980), 829–853.
- [2] M. Richardson and T. Smith. 1991. Tests of financial models in the presence of overlapping observations. *Rev. Financial Studies* 4, 2 (1991), 227–254.
- [3] J. Boudoukh, M. Richardson, and R. F. Whitelaw. 2008. The myth of long-horizon predictability. *Rev. Financial Studies* 21, 4 (2008), 1577–1605.
- [4] W. K. Newey and K. D. West. 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica* 55, 3 (1987), 703–708.
- [5] L. Kish. 1965. *Survey Sampling*. Wiley.
- [6] B. R. Moulton. 1990. An illustration of a pitfall in estimating the effects of aggregate variables on micro units. *Rev. Economics and Statistics* 72, 2 (1990), 334–338.
- [7] D. N. Politis and J. P. Romano. 1992. A circular block-resampling procedure for stationary data. In *Exploring the Limits of Bootstrap*. Wiley, 263–270.
- [8] M. H. Pesaran and A. Timmermann. 1992. A simple nonparametric test of predictive performance. *J. Business & Economic Statistics* 10, 4 (1992).
- [9] Y. Benjamini and Y. Hochberg. 1995. Controlling the false discovery rate. *J. Royal Statistical Society B* 57, 1 (1995), 289–300.
- [10] R. Sullivan, A. Timmermann, and H. White. 1999. Data-snooping, technical trading rule performance, and the bootstrap. *J. Finance* 54, 5 (1999), 1647–1691.
- [11] H. White. 2000. A reality check for data snooping. *Econometrica* 68, 5 (2000), 1097–1126.
- [12] P. R. Hansen. 2005. A test for superior predictive ability. *J. Business & Economic Statistics* 23, 4 (2005), 365–380.
- [13] C. R. Harvey, Y. Liu, and H. Zhu. 2016. ...and the cross-section of expected returns. *Rev. Financial Studies* 29, 1 (2016), 5–68.
- [14] D. H. Bailey and M. López de Prado. 2014. The deflated Sharpe ratio. *J. Portfolio Management* 40, 5 (2014), 94–107.
- [15] D. H. Bailey, J. Borwein, M. López de Prado, and Q. J. Zhu. 2017. The probability of backtest overfitting. *J. Computational Finance* 20, 4 (2017), 39–70.
- [16] M. López de Prado. 2018. *Advances in Financial Machine Learning*. Wiley.
- [17] A. Timmermann and C. W. J. Granger. 2004. Efficient market hypothesis and forecasting. *Int. J. Forecasting* 20, 1 (2004), 15–27.
- [18] R. F. Engle. 1982. Autoregressive conditional heteroscedasticity. *Econometrica* 50, 4 (1982), 987–1007.
- [19] T. Bollerslev. 1986. Generalized autoregressive conditional heteroskedasticity. *J. Econometrics* 31, 3 (1986), 307–327.
- [20] P. Carr and L. Wu. 2009. Variance risk premiums. *Rev. Financial Studies* 22, 3 (2009), 1311–1341.
- [21] G. Bakshi and N. Kapadia. 2003. Delta-hedged gains and the negative market volatility risk premium. *Rev. Financial Studies* 16, 2 (2003), 527–566.
- [22] P. N. Kolm, J. Turiel, and N. Westray. 2023. Deep order flow imbalance. *Mathematical Finance* 33, 4 (2023), 1044–1081.
- [23] Z. Zhang, S. Zohren, and S. Roberts. 2019. DeepLOB. *IEEE Trans. Signal Processing* 67, 11 (2019), 3001–3012.
- [24] S. Gu, B. Kelly, and D. Xiu. 2020. Empirical asset pricing via machine learning. *Rev. Financial Studies* 33, 5 (2020), 2223–2273.
- [25] T. Fischer and C. Krauss. 2018. Deep learning with LSTM networks for financial market predictions. *European J. Operational Research* 270, 2 (2018), 654–669.
- [26] C. Krauss, X. A. Do, and N. Huck. 2017. Deep neural networks, gradient-boosted trees, random forests. *European J. Operational Research* 259, 2 (2017), 689–702.
- [27] O. B. Sezer, M. U. Gudelek, and A. M. Ozbayoglu. 2020. Financial time series forecasting with deep learning. *Applied Soft Computing* 90 (2020), 106181.
- [28] B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister. 2021. Temporal fusion transformers. *Int. J. Forecasting* 37, 4 (2021), 1748–1764.
- [29] S. Hochreiter and J. Schmidhuber. 1997. Long short-term memory. *Neural Computation* 9, 8 (1997), 1735–1780.
- [30] A. Vaswani et al. 2017. Attention is all you need. In *NeurIPS*, Vol. 30, 5998–6008.
- [31] T. Chen and C. Guestrin. 2016. XGBoost. In *KDD*, 785–794.
- [32] G. Ke et al. 2017. LightGBM. In *NeurIPS*, Vol. 30, 3146–3154.
- [33] L. Breiman. 2001. Random forests. *Machine Learning* 45, 1 (2001), 5–32.
- [34] C. J. Clopper and E. S. Pearson. 1934. The use of confidence or fiducial limits. *Biometrika* 26, 4 (1934), 404–413.